

Concurrency and Distributed Systems

Week 8 – Part 4: Kafka Mental-Model Preview

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May 13, 2026

Part 4 Roadmap

1 Log Model

2 Bridge

Logs Are Ordered Facts

Mental model

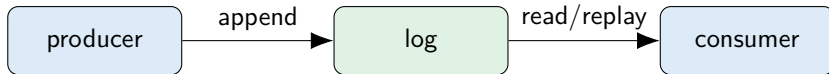
A log is an ordered history of things that happened.

- Not a queue only.
- Not a database table only.
- A replayable history that many consumers can read.

Why Logs Matter

Use	Meaning
History	What happened
Replay	Rebuild state by reading again
Recovery	Resume after failure
Audit	Explain past decisions
Decoupling	Producers and consumers move independently

Producer, Log, Consumer



Bridge From Concurrency

- BlockingQueue: local ordered handoff.
- CompletableFuture: async completion inside a process.
- Kafka-style log: ordered history across processes.
- The question changes from "who owns the lock?" to "what happened, in what order?"

Run

```
.\run-demo.ps1 05
```

- Name the five uses of logs.
- Connect replay to recovery.
- Connect decoupling to backpressure.